JAIDEV EDUCATION SOCIETY'S

J D COLLEGE OF ENGINEERING AND MANAGEMENT

KATOL ROAD, NAGPUR

(An Autonomous Institute, with NAAC "A" Grade)

Affiliated to DBATE, RTMNU & MSBTE Mumbai

Education to Eternity

Basic Science & Humanities Department

2023-24 (Semester)

CLASS TEST- I EXAMINATION

Course: B. Tech. (CSE/AI /DS)

Subject: NMCA

Subject Code:

Year/Semester: I[st] Semester (First Year)

Max Marks: 20 **Date: - 26/ 09/2023** **Duration: - 1 Hr**

Instructions to the Students:

1. All the questions carry marks as indicate.

2. Assume suitable data whenever necessary.

3. Illustrate your answer whenever necessary with the help of neat sketches.

CO1. Describe concept of linear algebra, probability, logic, numerical method, distribution theory.

CO2. Illustrate the concept of linear algebra, probability, logic, numerical method, distribution theory by using examples.

CO3. Apply the knowledge of linear algebra, probability, logic, numerical method, distribution theory to solve the engineering problems.

CO4. Analyze the problems and results of linear algebra, probability, logic, numerical method, distribution theory to solve the real-world problems.

CO5. Evaluate the problems by linear algebra, probability, logic, numerical method, distribution theory.

CO6. Create the methods or model by linear algebra, probability, logic, numerical method, distribution theory.

SECTION A

1. **All Questions are compulsory.**
2. **Answer in one word/sentence.**

Q. No.	Questions	Level/CO Mapped	Marks
Q.1	Write the formula for probability of any event A	1/CO1	1
Q.2	Define mutually exclusive events.	1/CO1	1
Q.3	Write down the formula for conditional probability of A given B.	1/CO1	1
Q.4	If A and B are any two events then write down the formula for P (A U B).	1/CO1	1

SECTION B

1. **All Questions are compulsory.**
2. **Solve answer in brief (two, three Sentence)**

Q.5	Find the sample space if two coins are tossed.	2/CO2	2
Q.6	If the probability of any event A is 0.6 then what will be the probability of A^c.	2/CO2	2
Q.7	State the formula for Baye's theorem.	2/CO2	2

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Q.8 Two cards are drawn from a pack of 52 cards. Find the probability that both are hearts. 3/CO3 2

SECTION C

1. **Solve any two questions.**

Q.9 An urn holds 5 white and 3 black marbles. If two marbles are drawn at random without replacement and X denotes the number of white marbles find (i) find probability function (ii) the distribution function. 4/CO4 4

Q.10 A random variable X has density function given by 3/CO3 4

$$f(x) = \begin{cases} k\,e^{-3x} & x > 0 \\ 0 & x \leq 0 \end{cases}$$

Find i) Constant k

ii) P (1 < x < 3)

Q.11 A discrete random variable x has following probability function 4/CO4 4

X	0	1	2	3	4	5	6	7
F(x)	0	k	2k	2k	3k	K^2	$2K^2$	$7K^2+k$

Find the value of K and P (x > 6)

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